Question	Active mass of 5 g CaO(s) is:
Type	multiple choice
Option	56
Option	1
Option	3.5
Option	2
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Ratio of active masses of 22g CO_2, 3g H_2 and 7g N_2 in a gaseous mixture is:
Type	multiple choice
Option	22:3:7
Option	0.5: 3:7
Option	1:3:1
Option	1:3: 0.5
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In a chemical equilibrium, the rate constant for the backward reaction is 7.5×10^{-4} and the equilibrium constant is 2. The rate constant for the forward reaction is:
Type	multiple choice
Option	2×10^{-3}
Option	1.12×10^{-4}
Option	1.5×10^{-3}
Option	9.0×10^{-6}
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	For which reaction is $K_P = K_C$
Type	multiple choice
Option	$2NOCl(g)=2NO(g)+Cl_2(g)$
Option	$N_2(g)+3H_2(g)=2NH_3(g)$
Option	$H_2(g)+I_2(g)\rightleftharpoons 2HI(g)$
Option	$2SO_2(g)+O_2(g)\rightleftharpoons 2SO_3(g)$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	At 1000 K, the value of K_P for the reaction $A(g)+2B\rightleftharpoons 3C(g)+D(g)$ is 0.05 atm. The value of Kc in terms of R would be:
Type	MULTIPLE CHOICE
Option	20000 R
Option	0.02 R
Option	$5\times10^{-5}R$
Option	$5\times10^{-5}\times R^{-1}$

Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	For which reaction at 298 K, the value of $\frac{K_P}{K_C}$ is maximum and minimum respectively $(A) N_2O_4 \rightleftharpoons 2NO_2$ $(B) 2SO_2 + O_2 \rightleftharpoons 2SO_3$ $(C) X + Y \rightleftharpoons 4Z$ $(D) A + 3B \rightleftharpoons 7C$
Type	multiple choice
Option	4, 3
Option	4, 2
Option	3, 2
Option	4, 1
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	For a reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the value of K_C does not depend upon: 1) Initial concentration of the reactants 2) Pressure 3) Temperature 4) Catalyst
Type	multiple choice
Option	Only 3
Option	1, 2 and 3
Option	1, 2 and 4
Option	1, 2, 3 and 4
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The equilibrium constant in a reversible reaction at a given temperature:
Type	multiple choice
Option	Depends on initial concentration of the reactants.
Option	Depends on the concentration of the products.
Option	Does not depend on the initial concentrations.
Option	It is not characteristic of the reaction.
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	For the reaction, $H_{2}(g) + I_{2}(g) \rightleftharpoons 2HI(g)$ equilibrium constant, K_{p} changes with:
Type	multiple choice
Option	Temperature
Option	Total pressure
Option	Catalyst
Option	Amount of H_{2} and I_{2} present

Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	For the reaction, $C(s)+CO_2(g) \rightleftharpoons 2CO(g)$ the partial pressure of CO & CO_2 are 2.0 & 4.0 atm respectively at equilibrium. The K_p for the reaction is
Type	multiple choice
Option	0.5
Option	4.0
Option	8.0
Option	1
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The equilibrium constant (K_p) for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g)+Cl_2(g)$ is 16. If the volume of the container is reduced to one-half its original volume, the value of K_p for the reaction at the same temperature will be:
Type	multiple choice
Option	32
Option	64
Option	16
Option	4
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	For the reaction, $P \rightleftharpoons Q+R$. Initially 2 mol of P was taken. Up to equilibrium 0.5 mol of P was dissociated. what would be the degree of dissociation:
Type	multiple choice
Option	0.5
Option	1
Option	0.25
Option	4.2
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	4 mole of PCl_5 are heated at constant temperature in closed container. If degree of dissociation for PCl_5 is 0.5 then calculate total number of moles at equilibrium
Type	multiple choice
Option	4.5
Option	6
Option	3
Option	4
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In a chemical equilibrium A +B $\rightleftharpoons$ C+ D when one mole each of the two

	reactants are mixed, 0.4 mol each of the products are formed. The equilibrium constant is:
Type	multiple choice
Option	1
Option	0.36
Option	2.25
Option	4/9
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In the reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2$ the partial pressure of PCl_3, Cl_2 and PCl_5 are 0.3, 0.2 and 0.6 atm respectively at equilibrium. If partial pressure of PCl_3 and Cl_2 was increased to twice, what will be the partial pressure of PCl_5 is in atm at new equilibrium condition:
Type	multiple choice
Option	0.3
Option	1.2
Option	2.4
Option	0.15
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In an equilibrium reaction for which $\Delta G^{\ominus} = 0$, the equilibrium constant K should be equal to:
Type	multiple choice
Option	Zero
Option	10
Option	1
Option	2
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The equilibrium constant K_p for the homogeneous gaseous reaction is 10^3. The standard Gibbs free energy change $\Delta G^{\ominus}$ for the reaction at 27°C (using R = 2 cal K-1 mol^{-1}) is
Type	multiple choice
Option	Zero
Option	-1.8 kcal
Option	-4.154 kcal
Option	+4.154 kcal
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	When 1 mole of pure ethyl alcohol (C_2H_5OH) is mixed with 1 mole of acetic acid at 25°C, the equilibrium mixture contains 2/3 mole each of ester and water $C_2H_5OH(l) + CH_3COOH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$ The $\Delta G^{\ominus}$ for the reaction at 298 K is:
Type	multiple choice

Option	3435 J
Option	4 J
Option	-3435 J
Option	Zero
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The value of $\Delta G^{\ominus}$ for a reaction in the aqueous phase having $K_C = 1$, would be:
Type	multiple choice
Option	-RT
Option	-1
Option	0
Option	+RT
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A plot of Gibbs energy of a reaction mixture against the extent of the reaction is:
Type	multiple choice
Option	minimum at equilibrium
Option	zero at equilibrium
Option	maximum at equilibrium
Option	none of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The equilibrium constant Kc for the reaction $P_4(g) \rightleftharpoons 2P_2(g)$ is 1.4 at 400°C. Suppose that 3 moles of $P_4(g)$ and 2 moles of $P_2(g)$ are mixed in 2 litre container at 400°C. What is the value of the reaction quotient (Qc)?
Type	multiple choice
Option	3/2
Option	2/3
Option	1
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	For the reaction, A(g) + 3B(g) $\rightleftharpoons$ 2C(g) at 27°C, 2 moles of A, 4 moles of B, and 6 moles of C are present in a 2-litre vessel. If Ke for the reaction is 1.2, the reaction will proceed in:
Type	multiple choice
Option	forward direction
Option	backward direction
Option	neither direction
Option	none of these
Answer	1
Solution	
Positive Marks	4

Negative Marks	1
Question	The standard free energy change of a reaction is $\Delta G^0 = -115\ kJ$ at 298 K. Calculate the value of $\log_{10} K_p$ ($R = 8.314\ JK^{-1}\ mol^{-1}$)
Type	multiple choice
Option	20.16
Option	2.303
Option	2.016
Option	13.83
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If the reaction quotient (Q) is greater than the equilibrium constant (K), what is true about the Gibbs free energy?
Type	multiple choice
Option	It is greater than zero
Option	It is less than zero
Option	More information is needed to determine the Gibbs free energy
Option	It is equal to zero
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Reaction quotient for the reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is given by, $A = \frac{[NH_3]^2}{[N_2][H_3]^2}$. The reaction will proceed from right to left if K_C value is:
Type	multiple choice
Option	$Q < K_C$
Option	$Q = 0$
Option	$Q > K_C$
Option	$Q = K_C$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Le Chatelier's principle is not applicable to:
Type	multiple choice
Option	$Fe(s) + S(s) \rightleftharpoons FeS(s)$
Option	$H_2(g) + I_2(g) = 2HI(g)$
Option	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3$
Option	$N_2(g) + O_2(g) = 2NO(g)$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Favorable conditions for the manufacturing of ammonia by the reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3; \Delta H = -21.9kcal$ are:
Type	multiple choice
Option	low temperature, low pressure and catalyst

Option	low temperature, high pressure and catalyst
Option	high temperature, low pressure and catalyst
Option	high temperature, high pressure and catalyst
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	According to Le Chatelier's principle adding heat to a solid and liquid in equilibrium with an endothermic nature will cause the:
Type	multiple choice
Option	amount of solid to decrease
Option	amount of liquid to decrease
Option	temperature to rise
Option	temperature to fall
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	In a vessel containing SO_3, SO_2 and O_2 at equilibrium some helium gas is introduced so that, the total pressure increases, while temperature and volume remain constant. According to Le Chatelier's principle the dissociation of SO_3 :
Type	multiple choice
Option	Increases
Option	Decreases
Option	remains unaltered
Option	changes unpredictably
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	For the reaction, $PCl_5(g) \rightleftharpoons PCl_3 + Cl_2(g)$ The forward reaction at constant temperature is favoured by:
Type	multiple choice
Option	introducing an inert gas at a constant volume
Option	introducing chlorine gas at constant volume
Option	introducing an inert gas at constant pressure
Option	none of the above
Answer	3
Solution	
Positive Marks	4
Negative Marks	1